

Car-following Models: Gipps Model

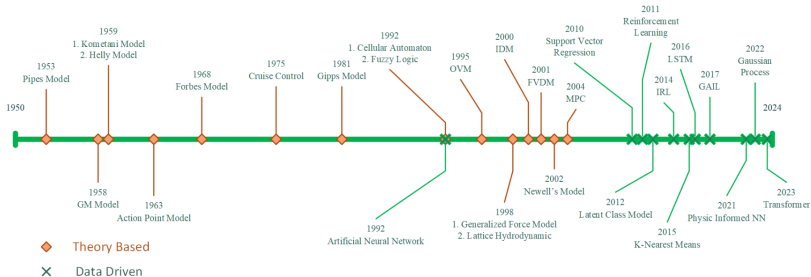
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<https://www.GoTrafficGo.com>

January 2, 2026

- Gipps, P. (1981). *A behavioural car-following model for computer simulation*. Transportation Research Part B
- The Gipps model is a **reaction-time-based** car-following model, i.e., The speed and location of a vehicle is updated for every **reaction time τ** , i.e. $v_i(t + \tau) = V(t)$
- To implement the Gipps model, the Runge-Kutta scheme is needed, because it is given in a discrete-time formulation of speed

Introduction



Tianya Zhang, et al., Car-Following Models: A Multidisciplinary Review, arXiv:2304.07143v, 2024

- The Gipps model was proposed in the discrete-time generic formulation. The **speed function** reads

$$v_i(t + \tau) = V[x_i(t), x_{i-1}(t), v_i(t), v_{i-1}(t)]$$

- Thus, we solve it by following

$$\boxed{\dot{x}_i(t) = v_i(t + \Delta t) = V(\cdot)} \xrightarrow{\text{RK}} \boxed{x_i(t + \Delta t)}$$

Combining the speed in **free-flow** and **congested** conditions, the speed function reads.

$$v_i(t + \tau) = \text{min} \left\{ v_i^{\text{free}}(t + \tau), v_i^{\text{cong}}(t + \tau) \right\}$$

It turns out the **smaller one** between the derived free-flow speed and congested speed.

Speed function in free-flow conditions

The speed in free-flow conditions is fitted from **empirical data** as

$$v_i^{\text{free}}(t + \tau) = v_i(t) + 2.5A_i\tau \left(1 - \frac{v_i(t)}{V_i^{\text{max}}}\right) \left(0.025 + \frac{v_i(t)}{V_i^{\text{max}}}\right)^{\frac{1}{2}}.$$

- V_i^{max} is the maximum speed;
- A_i is the acceleration.

Speed function in congested conditions

$v_i^{\text{cong}}(t)$ is proposed as follows

$$v_i^{\text{cong}}(t + \tau) = B_i \left(\frac{\tau}{2} + \theta \right) + \sqrt{B_i^2 \left(\frac{\tau}{2} + \theta \right)^2 + B_i \left\{ 2[x_{i-1}(t) - x_i(t) - L_{i-1}] - \tau v_i(t) - \frac{v_{i-1}(t)^2}{\hat{B}_{i-1}} \right\}}$$

- $B_i < 0$, deceleration;
- $\hat{B}_{i-1}$, braking rate of vehicle $(i - 1)$ estimated by vehicle i ;
- θ , safety margin time (talk later);
- L_i , vehicle length

Speed function in congested conditions

If we take $\theta = \frac{\tau}{2}$ as shown in [Wikipedia](#), the function (more recommended) is simplified as

$$v_i^{cong}(t + \tau) = B_i \tau + \sqrt{B_i^2 \tau^2 + B_i \left\{ 2[x_{i-1}(t) - x_i(t) - L_{i-1}] - \tau v_i(t) - \frac{v_{i-1}(t)^2}{\hat{B}_{i-1}} \right\}}$$

Part-1: Leading Vehicle

- If leading vehicle ($i - 1$) brakes **as hard as desirable** at time t , i.e., with deceleration B_{i-1} and speed **dropping to zero**, the vehicle will move to position x_{i-1}^*

$$x_{i-1}^* = x_{i-1}(t) + \frac{v_{i-1}(t)^2}{2B_{i-1}} \quad (1)$$

- WHY: Refer to the well-known uniform acceleration, the Equations of Motion are written as follows,

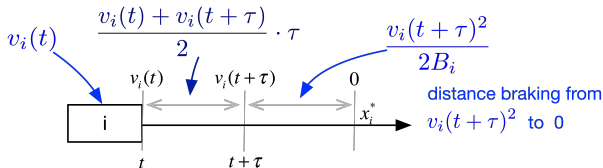
$$\begin{cases} v_t = v_0 + at \\ s = v_0 t + \frac{1}{2}at^2 \end{cases}$$

given v_0 (i.e., $v_{i-1}(t)$), a (i.e., B_{i-1}), calculate s

Part-2: Following Vehicle

- Following vehicle i will **not react until** time $(t + \tau)$, and will **stop** before reaching x_i^*

$$x_i^* = x_i(t) + \frac{v_i(t) + v_i(t + \tau)}{2} \cdot \tau + \frac{v_i(t + \tau)^2}{2B_i} \quad (2)$$



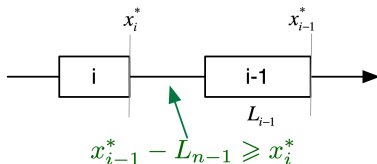
- The term, $\frac{v_i(t) + v_i(t + \tau)}{2} \cdot \tau$, is the distance that the vehicle moves within the reaction time τ .

It assumes **uniform acceleration**, also **second-order Runge-Kutta scheme**.

- For **safety reasons**, vehicle i must ensure that

$$x_{i-1}^* - L_{i-1} \geq x_i^* \quad (3)$$

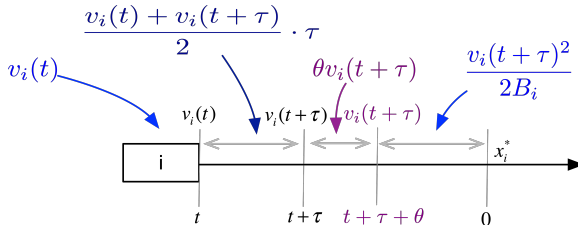
i.e., the rear bumper of the leader is in front of the front bumper of the follower.



Part-4: Human Error

- To allow the driver make mistakes, an **additional delay θ** is added. Thus, we have

$$x_{i-1}^* - L_{n-1} \geq x_i^* + \theta v_i(t + \tau) \quad (4)$$



Substituting Equation 1 and 2 into Inequality 4.

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$$x_i^* = v_i(t) + \frac{v_i(t) + v_i(t + \tau)}{2} \cdot \tau + \frac{v_i(t + \tau)^2}{2B_i} \quad (2)$$

Substituting Equation 1 and 2 into Inequality 4.

$$x_{i-1}^* = x_{i-1}(t) + \frac{v_{i-1}(t)^2}{2B_{i-1}} \quad (1)$$

$$\Downarrow$$

$$x_{i-1}^* - L_{i-1} \geq x_i^* + \theta v_i(t + \tau) \quad (4)$$

$$\Uparrow$$

$$x_i^* = v_i(t) + \frac{v_i(t) + v_i(t + \tau)}{2} \cdot \tau + \frac{v_i(t + \tau)^2}{2B_i} \quad (2)$$



$$x_{i-1}(t) + \frac{v_{i-1}(t)^2}{2B_{i-1}} - L_{i-1} \geq v_i(t) + \frac{v_i(t) + v_i(t + \tau)}{2} \cdot \tau + \frac{v_i(t + \tau)^2}{2B_i} + \theta v_i(t + \tau)$$

Substituting Equation 1 and 2 into Inequality 4.

$$x_{i-1}^* = x_{i-1}(t) + \frac{v_{i-1}(t)^2}{2B_{i-1}} \quad (1)$$

$$\Downarrow$$

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$$\Downarrow$$

$$x_{i-1}(t) + \frac{v_{i-1}(t)^2}{2B_{i-1}} - L_{i-1} \geq v_i(t) + \frac{v_i(t) + v_i(t + \tau)}{2} \cdot \tau + \frac{v_i(t + \tau)^2}{2B_i} + \theta v_i(t + \tau)$$

In real traffic, it is possible for the driver of vehicle i to estimate all the values in Equation 1 and 2 except B_{i-1} (i.e., deceleration of the leading vehicle) by direct observation.

Thus, B_{i-1} should be replaced by an estimate $\hat{B}_{i-1}$, and we finally obtain:

$$v_i(t + \tau) \leq B_i \left(\frac{\tau}{2} + \theta \right) + \sqrt{B_i^2 \left(\frac{\tau}{2} + \theta \right)^2 + B_i \left\{ 2[x_{i-1}(t) - x_i(t) - L_{i-1}] - \tau v_i(t) - \frac{v_{i-1}(t)^2}{\hat{B}_{i-1}} \right\}}$$

Wilson (2001) conducted simulation experiments on a ring road:

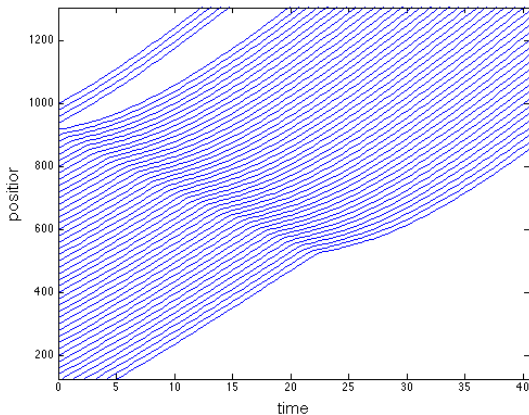
- In the experiment of “stable uniform flow”
 - uniformly distributed initial positions;
 - total vehicle number $N = 50$;
 - $A_i = 1.7 \text{ m/s}^2$;
 - $V_i^{\max} = 30 \text{ m/s}$;
 - $B_i = -3 \text{ m/s}^2$;
 - $\hat{B}_{i-1} = -3.5 \text{ m/s}^2$;
 - $\tau = 2/3$;
 - $\theta = 1/3$;
 - $L_i = 6.5 \text{ m}$.
- In the experiment of “unstable uniform flow, leading to a travelling wave”,
 - $\hat{B}_{i-1}$ is reduced to be 2.8 m/s^2

Prove that it is right: the performance of the model is consistent with reality **in the microscopic and macroscopic perspectives** (*that is why we need to know those important traffic characteristics*), such as

- (Simulated or analytically deducted) fundamental diagram
- Time-space of trajectories
- Stability analysis

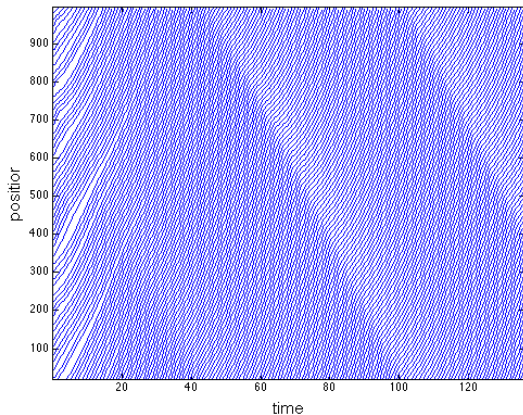
Thus, proposing a CF model usually needs a number of work.

Following works



Gipps model + straight road

Following works



Gipps model + ring road

Matlab codes

mg/OneDrive/Simulation/matlab/CF Model/Generic/A_1_Simulation.m

```
1 clear;
2 clc;
3
4
5
6 %%% Setting
7 global lengthOfRoad numberOfVehicle
8 lengthOfRoad=10000;
9 numberOfVehicle=400;
10
11 global lengthOfTimeStep
12 numberOfTimeStep=1000;
13 lengthOfTimeStep=0.5;
14
15 global percentOfInitialDifference
16 percentOfInitialDifference= [0 0.3];
17
18 global speedLimit
19 speedLimit= ConvertKMh2MS(100);
20
21 %%% Initialization
22 seed = RandStream('mt19937ar','Seed',111);
23 RandStream.setGlobalStream(seed);
24
25 global AllPosition AllSpeed
26 AllPosition= NaN(numberOfVehicle, numberOfTimeStep);
27 AllSpeed= NaN(numberOfVehicle, numberOfTimeStep);
28
29 %%%
30
31 for time_i=1:numberOfTimeStep
32
33     disp(time_i);
34
35     %%% load vehicles
36     if time_i==1
37
38         AllPosition(:,1)= SetInitialPosition(AllPosition(:,1));
39         AllSpeed(:,1)= randi([0,round(speedLimit)],numberOfVehicle,1);
40     % AllSpeed(:,1)= speedLimit;
41
42     %%% Simulation run
43     else
44
45         for veh_i=1:numberOfVehicle
46
47             newSpeed= UpdateSpeed(veh_i, time_i);
48             newPosition= UpdatePosition(veh_i, time_i, newSpeed);
49
50             AllPosition(veh_i, time_i)= newPosition;
51             AllSpeed(veh_i, time_i)= newSpeed;
52
53         end
54     end
55 end
56
57
58
59 %%%
60 save('SimulationData.mat');
```

mg/OneDrive/Simulation/matlab/CF Model/Generic/A_2_Plot.m

```
1 clc;
2 clear;
3
4
5 load('SimulationData.mat');
6
7 figure();
8
9 for veh_i=1:numberOfVehicle
10
11     indexStart=1;
12
13     loop=0;
14     for time_i=1:numberOfTimeStep-1
15
16         thisPosition= AllPosition(veh_i,time_i);
17         nextPosition= AllPosition(veh_i,time_i+1);
18
19         if thisPosition > nextPosition
20
21             %%%%%%%%%%%
22             loop=loop+1;
23             color= GetColor(loop);
24
25             indexEnd=time_i-1;
26
27             range= indexStart : 1 : indexEnd;
28             plot(range, AllPosition(veh_i, range), color);
29             hold on;
30             indexStart= time_i+1;
31
32         end
33     end
34
35     loop=loop+1;
36     color= GetColor(loop);
37
38     range= indexStart : 1 : numberOfTimeStep;
39     if size(range,2)>1
40         plot(range, AllPosition(veh_i, range), color); hold on;
41     end
42
43     set(gcf,'Position',[0,0,1500,800])
44 end
```

Thank you!